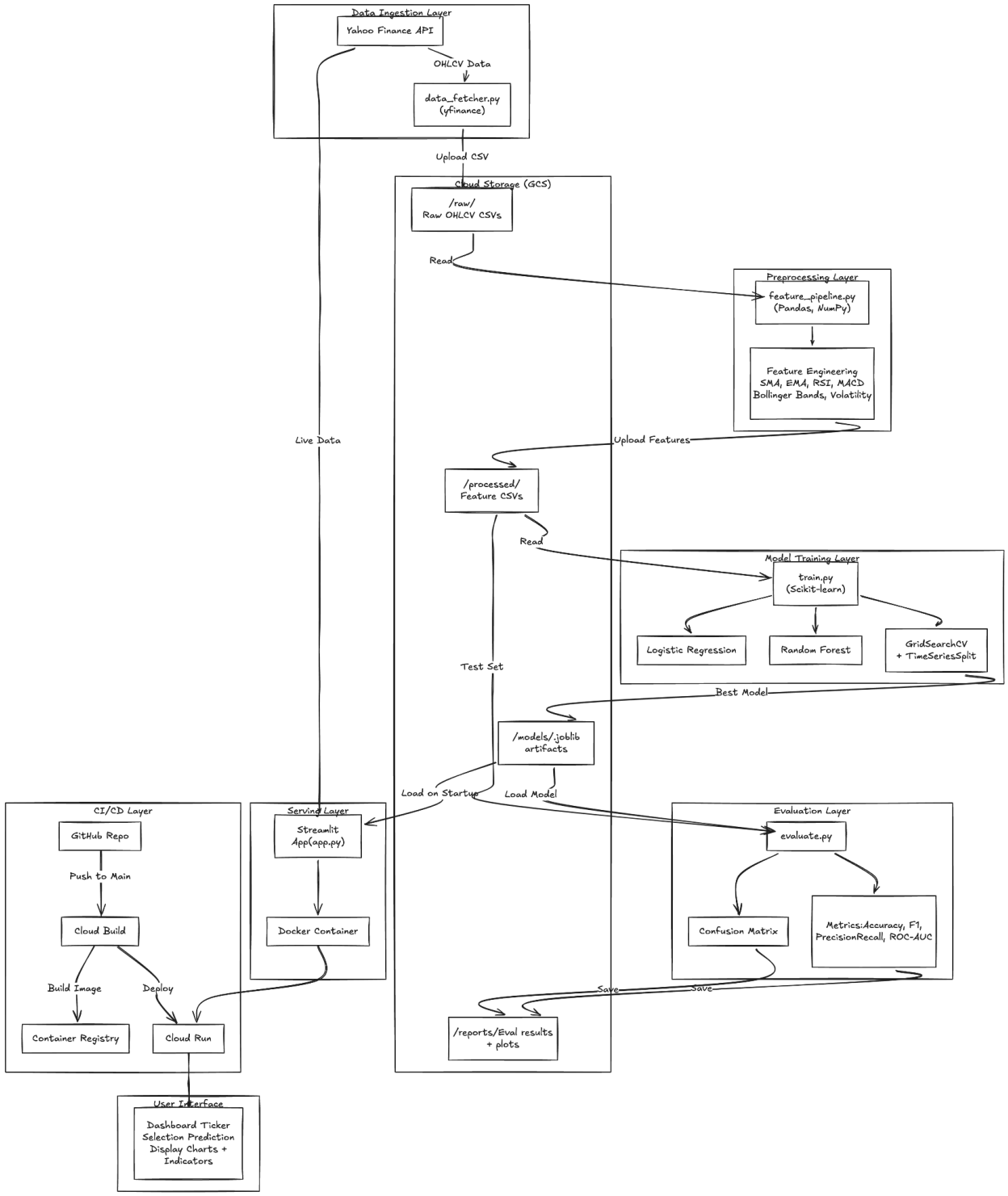
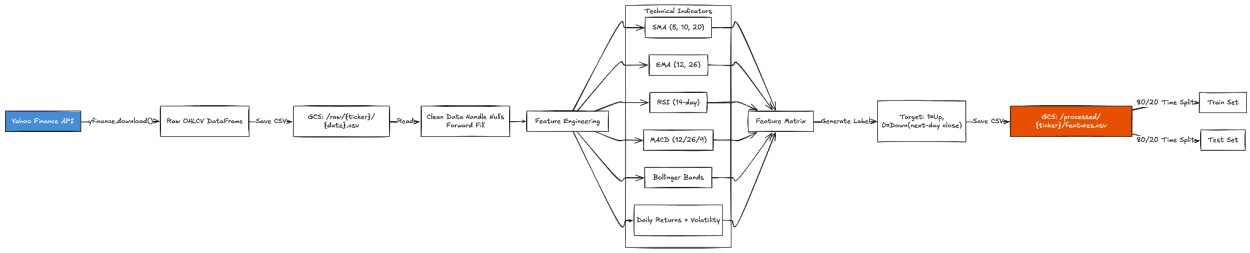
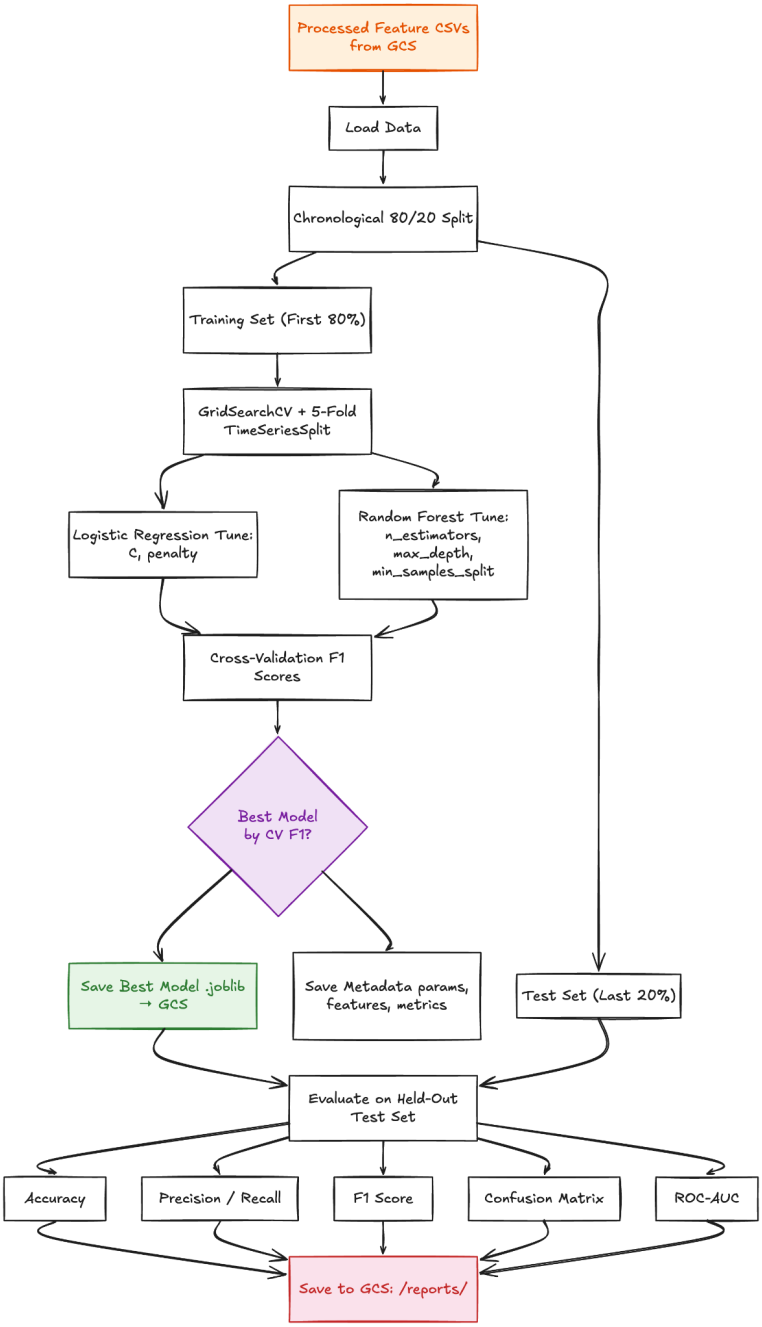
Full system architecture



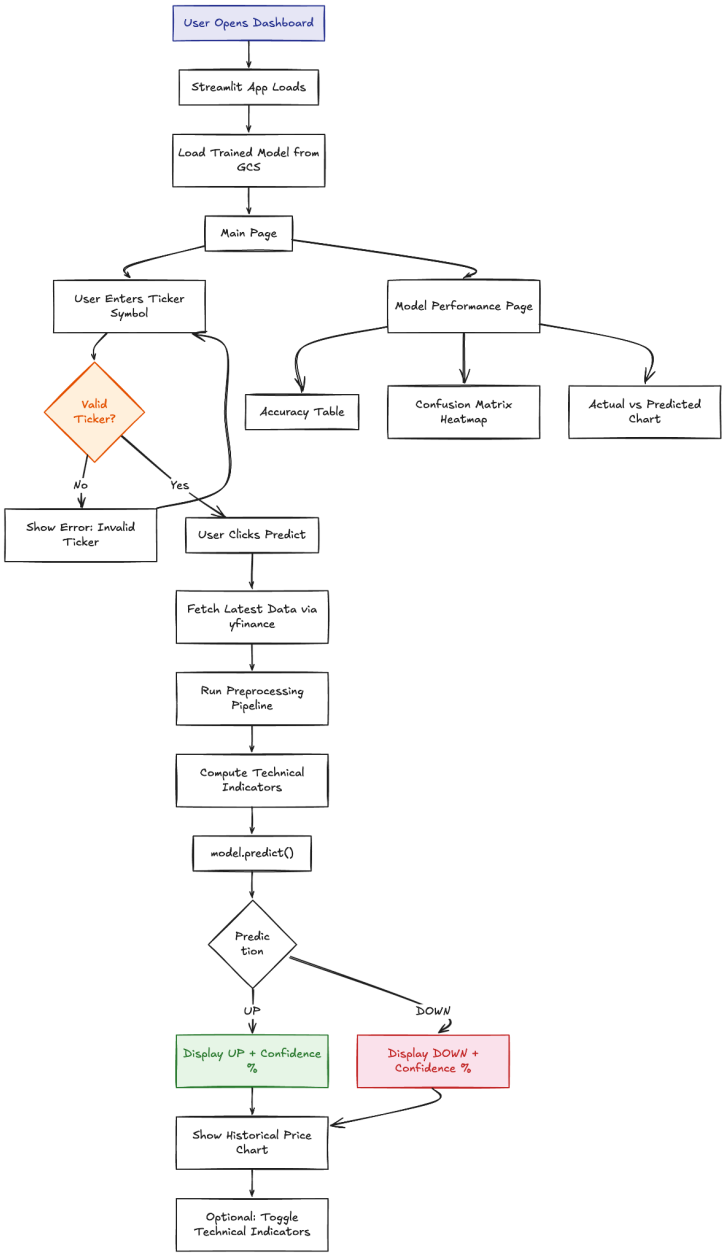
Data pipeline flow



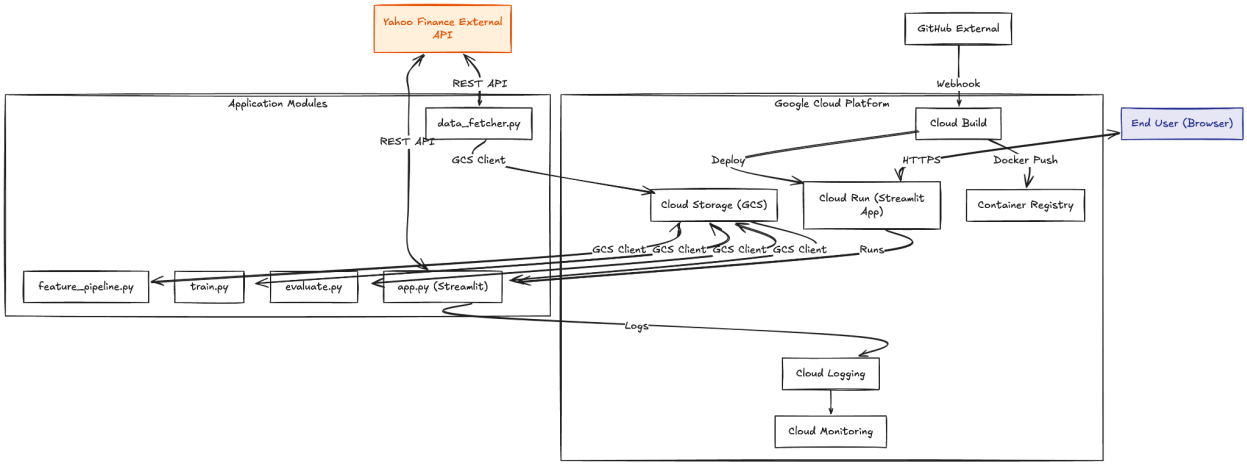
Training pipeline



User interaction flow



Component communication map


Yahoo Finance External API
REST API
Application Modules
data_fetcher.py
REST API
GCS Client
feature_pipeline.py
train.py
evaluate.py
app.py (Streamlit)
GitHub External
Webhook
Google Cloud Platform
Cloud Build
End User (Browser)
Deploy
HTTPS
Docker Push
Cloud Storage (GCS)
Cloud Run (Streamlit App)
Container Registry
GCS Client
GCS Client
GCS Client
GCS Client
Runs
Logs
Cloud Logging
Cloud Monitoring